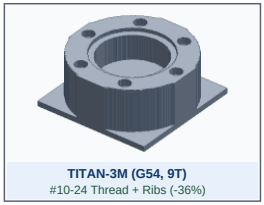
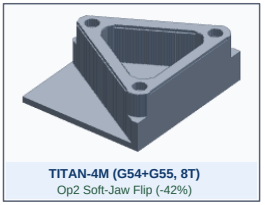
## (a) Automated ISO 10303-21 .step -> Carved Stock



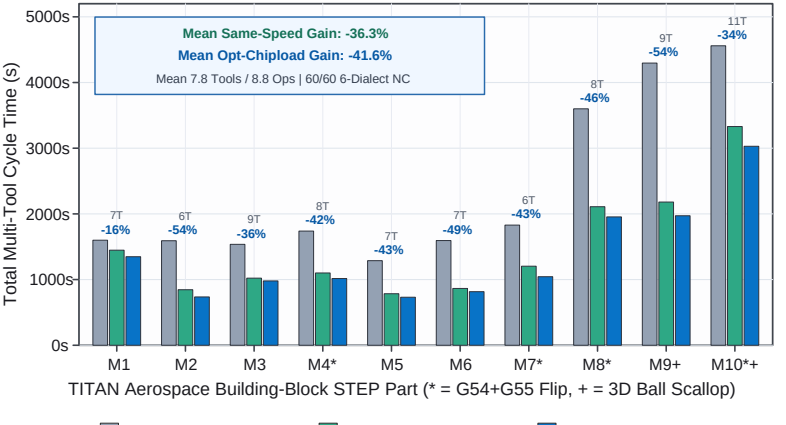
**TITAN-3M (G54, 9T)**
#10-24 Thread + Ribs (-36%)



**TITAN-4M (G54+G55, 8T)**
Op2 Soft-Jaw Flip (-42%)

**TITAN-9M (G54, 9T)**
3D B-Spline Dome (-54%)

**TITAN-10M (G54+G55, 11T)**
3D Airfoil + 11 Thr (-34%)

**All 10/10 STEP parts: 0.000 mm3 gouge | 0 vise/holder/rapid collisions**

## (b) 10-Part TITAN STEP Cycle Time: Naive vs. Same-Speed vs. Opt-Chipload